Chandan Sankar

Minneapolis, MN | +1 (612) 399-6938 | sankar.chandan@gmail.com
jahndan.cc | in/jahndan | jahndan

Education

University of Minnesota–Twin Cities *September 2022—(Present)*

- Master of Science in Electrical and Computer Engineering (Expected 2027)
- Bachelor of Computer Engineering, Bachelor of Science in Computer Science, Minor in Mathematics (Expected 2026)
- Cumulative GPA: 3.670, Officer of IEEE Student Chapter, Co-founder of ECE Modsynth, Member of Social Coding

Experience

Samsung SmartThings – *Software Engineering Intern (Hub Platform Team)* *June 2025—August 2025*

- Prototyped new "over-the-air" upgrade flow for Zigbee IoT devices, reducing standalone Hub firmware sizes by up to 35% to improve storage, update speed, and network efficiency.
- Developed Bash/Python tools to profile hubs, benchmark performance, and detect firmware regressions.
- Gained exposure to IoT networking protocols and system-level debugging of embedded Linux hubs.

Projects

Path Tracer – *Physically based 3D software renderer* *November 2025—December 2025*

- Implemented a Monte Carlo path tracer with task parallelization using OpenMP and data parallelization (SIMD) using glm.
- Simulated global illumination effects including ambient occlusion, reflection, refraction, depth of field, and area lighting.
- Designed an efficient ray intersection pipeline supporting geometric primitives, accelerated with a bounding volume hierarchy.

ECE Modsynth – *Modular synthesizer from scratch* *November 2024—May 2025*

- Co-founded department-backed project: securing funding, recruiting 5 team members, and developing introductory training materials including interactive synthesizer demos to explain modular synthesis.
- Prototyped phase-accurate oscillator with frequency modulation capabilities on an RP2040 with a hand-rolled fixed-point math library, enabling real-time DSP for hi-fi audio under constrained hardware resources.
- Debugged oscillator and pulse-generation circuits using an oscilloscope to verify waveform accuracy and timing margins under hardware constraints.

UMN Course Flowchart – *Course prerequisite visualizer webapp* *September 2023—January 2025*

- Co-founded project and led web app team, recruited 7 new members to round out web app team for front-end development.
- Proposed intuitive visual representation of nested prerequisites (which have any/all sub-requirements).
- Formulated digraph-building algorithm that traverses courses' prerequisites, with an optimized layout to improve readability.

Leadership

IEEE Student Chapter at UMN – *Officer (Secretary)* *May 2025—(Present)*

- Delivered lecture on foundational computer engineering concepts like abstraction, building up from logic gates to a computer.
- Streamlined operations by maintaining meeting documentation and developing automations to reduce manual overhead.
- Replaced the university CMS with a statically generated site under version control, improving maintainability and simplifying chapter-wide communications such as newsletters, slides, and announcements.
- Co-led technical committee project, developing a TinyUSB-based "plug-and-play" game controller to demonstrate embedded systems and digital interface design.

Skills & Coursework

Adaptive Digital Signal Processing	Algorithms and Data Structures	C	Verilog	Vivado	GNU Make
Fundamentals of Computer Graphics	Advanced Computer Architecture	C++	Typescript	ReactJS	MATLAB
Statistical Learning and Inference	Analysis of Numerical Algorithms	Rust	Javascript	AstroJS	Simulink
Linear Control Systems	Applied Linear Algebra	Python	Java	OpenGL	KiCad
Digital Filter Design	Introductory Abstract Algebra	Bash	Ocaml	Git	Oscilloscope